



PAPER – 4: COST AND MANAGEMENT ACCOUNTING



QUESTIONS

PART I - Case Scenario based MCQs

Standard Costing

1. ALZO Toys Ltd. is an exciting new player in the toy manufacturing industry, founded with a passion for creating high-quality, engaging, and educational toys. The company aims to make a positive impact on the industry and also on the development of young minds through imaginative play.

The following statement provides a comprehensive analysis of the various cost variances for a particular period, outlining the differences between the expected costs and the actual expenditures incurred.

Cost variances	(₹)
Direct material price	25,000F
Direct material usage	3,750A
Direct labour rate	5,000A
Direct labour efficiency	3,750A
Variable overhead expenditure	15,000A
Variable overhead efficiency	1,875A
Fixed overhead expenditure	62,500F

The budget for the same period reflected the following data:

Production volume	7,500 units
Direct materials purchased	3,750kg
Direct materials used	3,750kg
Direct material cost	₹ 1,12,500
Direct labour hours	5,625 hours
Direct labour cost	₹ 1,12,500
Variable overhead cost	₹ 56,250
Fixed overhead cost	₹ 1,12,500

Some other information relating to the same period is provided below:

- (i) Stocks of raw materials and finished goods are valued at a predetermined standard cost for easier cost comparison and reporting.
- (ii) The actual number of units produced was 7,750.
- (iii) The direct materials purchased were 5,000 kg.

From the information given above, you are required to FIGURE OUT the following in actual:

- (i) Quantity of materials used and direct material cost-
 - (a) 3,875 kg and ₹ 1,50,000
 - (b) 3,875 kg and ₹ 1,25,000
 - (c) 4,000 kg and ₹ 1,25,000
 - (d) 4,000 kg and ₹ 1,50,000
- (ii) Direct labour hours-
 - (a) 7,937.50 hours
 - (b) 6,000 hours
 - (c) 5,812.50 hours
 - (d) 5,000 hours

- (iii) Direct labour cost-
 - (a) ₹ 1,16,250
 - (b) ₹ 1,25,000
 - (c) ₹ 1,55,000
 - (d) ₹ 1,63,750
- (iv) Variable overhead cost-
 - (a) ₹ 75,000
 - (b) ₹ 73,125
 - (c) ₹ 60,000
 - (d) ₹ 58,125
- (v) Fixed overhead cost-
 - (a) ₹ 1,75,000
 - (b) ₹ 1,12,500
 - (c) ₹ 62,500
 - (d) ₹ 50,000

Activity based Costing

2. With the rise in carbon dioxide, a greenhouse gas, release in the environment, when fossil fuels are burned, Earth's natural greenhouse effect is becoming too weak causing global warming.

To contribute towards global environment for the betterment, various cars and scooters manufacturing companies are shifting their production towards electric vehicles (EVs) manufacturing. Companies are heavily investing in research, development and production of EVs.

Olay Ltd. is also one the companies belonging to scooter manufacturing industry. Watching its rivalries shifting their production to EVs, the management of Olay Ltd. decided to take the advantage of this open opportunity.

It had been only 4 years since the company started its production of EVs, but last year, the management of the company also decided to expand

its product line to 3 variants of the scooter, viz., Olay EV Max, Olay EV Ultra and Olay EV Pro.

The following information is provided for the current year from the books of Olay Ltd.:

	Olay EV Max	Olay EV Ultra	Olay EV Pro
Average revenue per unit (₹)	84,975	1,15,500	2,17,800
Average cost of goods sold per unit (₹)	82,500	1,10,000	1,98,000

Last year, when the company initially expanded its product line, the sales were not much noticeable comparative to its earlier category of scooters. The company could only sold 528, 330 and 110 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively against the order received of 616, 396 and 165 units respectively. However, with the increasing awareness about the EVs, more people started buying EVs and, during the current year, the company's order and sales jumped to five times the last year.

For earlier years, Olay Ltd. used gross margin percentage method to evaluate the relative profitability for all of its EVs.

However, from current year, company plans to use activity based costing for analysing the profitability.

The Activity analysis of Olay Ltd. is as under:

Activity Area	Cost-allocation base
Customer purchase order processing	Purchase orders by customers
Line-item ordering	Line-items per purchase order
Store delivery	Unit sold
Cartons dispatched to stores	Cartons dispatched to a store per Unit
Shelf-stocking at customer store	Hours of shelf-stocking

All the support costs for the current year amounts to ₹ 66,23,760. These support costs are assigned to all the activity areas. The cost in each area

and the quantity of the cost allocation basis used in that area are as follows:

Activity Area	Total costs (₹)	Total Units of Cost-allocation base
Line-item ordering	14,04,480	66,110 line items
Cartons dispatched to store	16,72,000	3,39,240 cartons
Shelf-stocking at customer store	2,25,280	2,904 hours

The Customer purchase order processing costs ₹ 17,60,000 along with the store delivery cost of ₹ 15,62,000.

Some of the other information is also provided below:

	Olay EV Max	Olay EV Ultra	Olay EV Pro
Average number of line items per order	10	12	14
Average number of cartons shipped per store unit	16	80	300
Average number of hours of shelf-stocking per store delivery	0.1	0.6	3

The company wants you to FIGURE OUT the following to ascertain which of the product line is more profitable:

- (i) For the current year, how much is the order received and the units sold for Olay EV Max, Olay EV Ultra and Olay EV Pro respectively?
 - (a) Order received- 616, 396 and 165 units; sold- 528, 330 and 110 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.
 - (b) Order received- 3,696, 2,376 and 990 units; sold- 3,168, 1,980 and 660 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.
 - (c) Order received- 528, 330 and 110 units; sold- 616, 396 and 165 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.

- (d) Order received- 3,080, 1,980 and 825 units; sold- 2,640, 1,650 and 550 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.
- (ii) The total gross-margin percentage and the operating income percentage, for the current year would be:
- (a) Gross-margin - 3.72% and Operating income - 4.96%
 - (b) Gross-margin - 4.96% and Operating income - 3.72%
 - (c) Gross-margin - 4.96% and Operating income - 4.96%
 - (d) Gross-margin - 3.72% and Operating income - 3.72%
- (iii) The cost driver rate relating to all the five activity areas would be:
- (a) Customer purchase order processing- ₹ 363.64 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 265.42 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.
 - (b) Customer purchase order processing- ₹ 299.07 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 322.73 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.
 - (c) Customer purchase order processing- ₹ 299.07 per order, Line item ordering- ₹ 77.58 per line item order, Store delivery- ₹ 322.73 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 21.24 per hour.
 - (d) Customer purchase order processing- ₹ 322.73 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 299.07 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.

- (iv) The operating cost of the individual product line, as per the method proposed for the current year w.r.t. profitability analysis, would be:
- Olay EV Max- ₹ 16,11,013, Olay EV Ultra- ₹ 23,56,890 and Olay EV Pro- ₹ 26,56,059
 - Olay EV Max- ₹ 23,56,890, Olay EV Ultra- ₹ 26,56,059 and Olay EV Pro- ₹ 16,11,013
 - Olay EV Max- ₹ 26,56,059, Olay EV Ultra- ₹ 16,11,013 and Olay EV Pro- ₹ 23,56,890
 - Olay EV Max- ₹ 26,56,059, Olay EV Ultra- ₹ 23,56,890 and Olay EV Pro- ₹ 16,11,013
- (v) Operating income as a percentage of revenues of each product line, namely Olay EV Max, Olay EV Ultra and Olay EV Pro, when all the support costs are allocated using an activity-based costing system would be:
- Olay EV Max- 1.73%, Olay EV Ultra- 3.53% and Olay EV Pro- 7.75%
 - Olay EV Max- 1.18%, Olay EV Ultra- 1.24% and Olay EV Pro- 1.34%
 - Olay EV Max- 2.91%, Olay EV Ultra- 4.76% and Olay EV Pro- 9.09%
 - Olay EV Max- 1.78%, Olay EV Ultra- 3.70% and Olay EV Pro- 8.52%

Employee Cost and Direct Expenses

3. Suppose the units of a product are produced at the rate of 4 units per useful direct labour hour. Direct labour idle time is 10% of hours paid for. Sales of 1,000 units are budgeted and finished goods stock is expected to rise by 100 units. The budgeted direct labour hours for the production would be:
- 306
 - 250

- (c) 275
- (d) 400

Cost Accounting Systems

4. WHICH of the following is the correct journal entry as would appear in the cost books when overhead expenses incurred ₹ 500 (Production ₹ 250 and Administrative ₹ 250)?

(a)	Production Overhead Control A/c	Dr.	₹ 250	
	Administrative Overhead Control A/c	Dr.	₹ 250	
	To Cost Ledger Control A/c			₹ 500
(b)	Production Overhead Control A/c	Dr.	₹ 250	
	Administrative Overhead Control A/c	Dr.	₹ 250	
	To Overhead Expenses A/c			₹ 500
(c)	Cost Ledger Control A/c	Dr.	₹ 500	
	To Production Overhead Control A/c			₹ 250
	To Administrative Overhead Control A/c			₹ 250
(d)	Overhead Expenses A/c	Dr.	₹ 500	
	To Production Overhead Control A/c			₹ 250
	To Administrative Overhead Control A/c			₹ 250

Joint Products and by products

5. In a company's production process, two joint products, A and B, are created simultaneously. The company operates in the manufacturing sector, focusing on producing goods that require multiple stages of production, often resulting in the creation of joint products that share common production costs up until a certain split-off point. For the last month, the following information is provided relating to the inventory and sales:

Product	Sales (units)	Finished Goods (units)	
		Opening stock	Closing stock
A	1,14,000	1,900	5,700
B	76,000	7,600	3,800

Joint production costs for the last month amounted to ₹ 20,90,000. These costs were allocated between both the joint products, A and B, based on the number of units produced. This method of apportioning costs ensures that each product is fairly charged for its share of the overall production expenses.

CALCULATE the joint production costs apportioned to product A for last month?

- (a) ₹ 12,12,200
- (b) ₹ 12,54,000
- (c) ₹ 12,95,800
- (d) ₹ 13,58,500

Marginal Costing

6. A company's sales could decrease by 50% before it starts incurring losses. And, for every Re. 1 of sales, it can contribute 40 paise towards fixed costs and generating profit. For the current year, the company's fixed cost amounts to ₹ 5,00,000.

CALCULATE the Projected sales.

- (a) ₹ 10,00,000
- (b) ₹ 12,50,000
- (c) ₹ 20,83,333
- (d) ₹ 25,00,000

PART - II Descriptive Questions**Material Cost**

7. XYZ Ltd is engaged in the business of manufacturing product "A". The material required for the manufacture of Product A is Material "B". Each unit of Product A requires 2 Kgs of Material B, however the yield of each unit of Material B is 80% of the input.

The Details governing the production requirements for the year is as follows:

- | | | | |
|----|---|---|---------------|
| a. | Projected sales of product A for the period | - | 10,000 units. |
| b. | Opening stock of product A | - | 948 units |
| c. | Opening stock of Raw Material B | - | 2,630 Kgs |
| d. | Ordering cost per order | - | ₹500 |
| e. | Rate of Overdraft | - | 12.5% |
| f. | Obsolescence Rate | - | 12.5% |
| g. | Purchase price of Material B | - | ₹ 20 per Kg |
| h. | Current purchase policy | - | Quarterly |
| i. | Max usage is more than minimum usage by | - | 100 Kgs |
| j. | Waiting time from date of order | - | 4 to 8 days |

Based on the above information determine the following:

- i. Economic Order Quantity
- ii. Re-Order Quantity
- iii. Re-Order Level
- iv. Maximum Stock
- v. Minimum Stock
- vi. Profitability analysis on EOQ v. ROQ
- vii. If a volume discount of 5% is provided for a quarterly purchase by the supplier, will your answer to (vi) above change?

Employee Cost and Direct Expenses

8. The existing incentive system of Sakshi Limited is as under:

Normal working week	5 days of 8 hours each plus 3 late shifts of 3 hours each
Rate of Payment	Day work: ₹ 160 per hour Late shift: ₹ 225 per hour
Average output per operator for 49-hours week i.e. including 3 late shifts	240 articles

In order to increase output and eliminate overtime, it is decided to switch on to a system of payment by results. The following information is obtained:

Time-rate (as usual)	: ₹ 160 per hour
Basic time allowed for 15 articles	: 2.5 hours
Piece-work rate	: Add 20% to basic piece-rate
Premium Bonus	: Add 50% to time.

If during the last week 270 articles are produced in a 40-hour week.

Required:

- (i) CALCULATE weekly earnings for one operator under the following systems:
 - (a) Existing time-rate
 - (b) Straight piece-work
 - (c) Rowan system
 - (d) Halsey premium system
- (ii) PREPARE a Statement showing hours worked, weekly earnings, number of articles produced and labour cost per article for one operator under the above systems.

Overheads- Absorption Costing Method

9. SA Ltd. has three production (M_1 , M_2 and A_1) and three service departments (Stores, Engineering services and General service). Engineering department serves the M_1 and M_2 only.

The relevant information related with Product X and Y are as follows:

	Product X	Product Y
M_1	10 Machine hours	6 Machine hours
M_2	4 Machine hours	14 Machine hours
A_1	14 Direct labour hours	18 Direct labour hours

The annual budgeted overhead cost for the year is

	Indirect Wages (₹)	Consumable Supplies (₹)
M_1	9,30,400	2,52,000
M_2	8,26,800	3,64,000
A_1	3,24,400	84,000
Stores	1,64,000	56,000
Engineering Service	1,06,800	84,000
General Service	1,50,400	64,000

	(₹)	
– Depreciation on Machinery	7,92,000	
– Insurance of Machinery	1,44,000	
– Insurance of Building	64,800	(Total building insurance cost for M_1 is one third of annual premium)
– Power	1,29,600	
– Light	1,08,000	
– Rent	2,53,500	(The general service deptt. is located in a

		building owned by the company. It is valued at ₹ 1,20,000 and is charged into cost at notional value of 8% per annum. This cost is additional to the rent shown above)
The value of issues of materials to the production departments are in the same proportion as shown above for the Consumable supplies.		

The following data are also available:

Department	Book value Machinery (₹)	Area (Sq. ft.)	Effective H.P. hours %	Production Direct Labour hour	Capacity Machine hour
M ₁	24,00,000	5,000	50	2,00,000	40,000
M ₂	18,00,000	6,000	35	1,50,000	50,000
A ₁	6,00,000	8,000	05	3,00,000	—
Stores	2,40,000	2,000	—	—	—
Engg. Service	7,20,000	2,500	10	—	—
General Service	2,40,000	1,500	—	—	—

Required:

- PREPARE an overhead analysis sheet, showing the bases of apportionment of overhead to departments.
- PREPARE a statement allocating service department overheads to production department ignoring the apportionment of service department costs among service departments.
- CALCULATE suitable overhead absorption rate for the production departments.
- CALCULATE the overheads to be absorbed by two products, X and Y.

Cost Sheet

10. The accounts of Ranu Ltd for the year ended 31st March, 2025, shows the following:

Particulars	(₹)
Carriage Inward	7,150
Administrative Office Salaries	12,600
Bad debts written off	6,500
Carriage Outward	4,300
GST (ITC Allowed)	10,000
Salaries related to factory office	7,000
Production Wages	1,20,000
Repairs - Plant and Machinery	4,450
Rent, Rates, Taxes, Insurance etc.	
Factory	8,500
Office	2,000
Sales	4,50,000
Stock of Raw materials:	
1 st April, 2024	50,000
31 st March, 2025	64,500
Materials Purchased (excluding GST)	2,00,000
Travelling Expenses	2,100
Manager's Salary (1/4 Office and 3/4 Factory)	10,000
Depreciation on Plant and Machinery	6,500
Depreciation on Office Furniture	500
Director's Fees	6,000
Gas and Water (Factory)	1,500
Gas and Water (Office)	500
General Expenses	3,400

You are required to PREPARE a cost statement for the year ended 31st March, 2025.

Cost Accounting Systems

11. The following figures have been taken from the financial accounts of a manufacturing firm for the year ended 31st March, 2025:

	(₹)
Direct material consumption	20,00,000
Direct wages	12,00,000
Factory overheads	6,40,000
Repair and maintenance (Factory-related)	48,000
Advertisement expenses (for brand building – capital in nature)	40,000
Administrative overheads	2,80,000
Selling and distribution overheads	3,84,000
Bad debts	32,000
Preliminary expenses written off	16,000
Legal charges	4,000
Depreciation on Plant and Machinery	1,20,000
Interest on Loan	60,000
Dividends received	40,000
Interest on fixed deposit	8,000
Sales- 48,000 units	48,00,000
Closing stock:	
- Finished stock- 4,000 units	3,20,000
- Work-in-process	96,000

The cost accounts for the same period reveal that the direct material consumption was ₹ 22,40,000; Factory overhead (including factory related repairs and maintenance and depreciation) is recovered at 20% on prime cost; Administration overhead is recovered @ ₹ 4.8 per unit of

cost of goods sold and selling and distribution overheads are recovered at ₹6.40 per unit sold.

Required:

PREPARE Costing and Financial Profit & Loss Accounts and RECONCILE the difference in the profit as arrived at in the two sets of accounts.

Job and Batch Costing

12. Pinku Confectioners (PC) owns a bakery which is used to make bakery items like pastries, cakes and muffins. PC use to bake atleast 50 units of any item at a time. A customer has given an order for 600 cakes. To process a batch of 50 cakes, the following cost would be incurred:

Direct materials- ₹ 5,000

Direct wages- ₹ 500

Oven set- up cost ₹ 750

PC absorbs production overheads at a rate of 20% of direct wages cost. 10% is added to the total production cost of each batch to allow for selling, distribution and administration overheads.

PC requires a profit margin of 25% of sales value.

Required:

- (i) Determine the price to be charged for 600 cakes.
- (ii) Calculate cost and selling price per cake.
- (iii) What would be selling price per unit If the order is for 605 cakes.

Process & Operation Costing

13. A product passes through three processes – A, B and C. The details of expenses incurred on the three processes during the year 2025 were as under:

Process	A	B	C
Units issued / introduced cost per unit ₹ 100	10,000		

	₹	₹	₹
Sundry Materials	10,000	15,000	5,000
Labour	30,000	80,000	65,000
Direct Expenses	6,000	18,150	27,200
Selling price per unit of output	120	165	250

Management expenses during the year were ₹ 80,000 and selling expenses were ₹ 50,000 these are not allocable to the processes.

Actual output of the three processes was:

A – 9,300 units, B-5,400 units and C-2,100 units. Two third of the output of Process A and one half of the output of Process B was passed on to the next process and the balance was sold. The entire output of process C was sold.

The normal loss of the three processes, calculated on the input of every process was:

Process A-5%; B-15% and C-20%

The Loss of Process A was sold at ₹ 2 per unit, that of B at ₹ 5 per unit and of Process C at ₹ 10 per unit.

Prepare the Three Processes Accounts and the Profit and Loss Account.

Joint Products and By products

14. A company processes a raw material in its Department 1 to produce three products, viz. A, B and X at the same split-off stage. During a period 1,80,000 kgs of raw materials were processed in Department 1 at a total cost of ₹ 12,88,000 and the resultant output of A, B and X were 18,000 kgs, 10,000 kgs and 54,000 kgs respectively. A and B were further processed in Department 2 at a cost of ₹ 1,80,000 and ₹ 1,50,000 respectively.

X was further processed in Department 3 at a cost of ₹1,08,000. There is no waste in further processing. The details of sales affected during the period were as under:

	A	B	X
Quantity Sold (kgs.)	17,000	5,000	44,000
Sales Value (₹)	12,24,000	2,50,000	7,92,000

There were no opening stocks. If these products were sold at split-off stage, the selling prices of A, B and X would have been ₹ 50, ₹ 40 and ₹ 10 per kg respectively. Required:

- (i) Prepare a statement showing the apportionment of joint costs to A, B and X.
- (ii) Present a statement showing the cost per kg of each product indicating joint cost and further processing cost and total cost separately.
- (iii) Prepare a statement showing the product wise and total profit for the period.
- (iv) State with supporting calculations as to whether any or all the products should be further processed or not

Service Costing

15. Due to frequent power cuts and inconsistent water supply from the Electricity Board, Anju Manufacturing Pvt. Ltd. — a mid-sized engineering goods manufacturer based in Pune — decided to set up its own captive power generation plant. The plant is critical for ensuring uninterrupted production across its two major units:
 - Unit 1: Produces industrial machinery components (highly power-intensive operations)
 - Unit 2: Manufactures agricultural equipment parts (moderate power load)

The factory operations heavily depend on continuous electric and water supply for running CNC machines, foundry operations, welding units, and assembly lines. Any electricity failure leads to major production delays, idle labour costs, and penalties from clients due to late deliveries.

After analysing the risk of depending entirely on the electricity board, the company's Board of Directors approved a proposal to invest in a

dedicated steam boiler and power generation plant within the factory premises.

The following information has been taken from the records in connection with the generation of power during the month:

- Coal used 1,800 Tonnes @ ₹ 230 per tonne
- Freight and handling charges: 7% of the value of coal used.
- Oil: 15 Tonnes @ ₹ 4,500 per tonne
- Water: 80,000 litres @ ₹ 0.07 per litre
- Steam boiler cost ₹ 50,00,000 with a residual value of ₹ 2 lakh after a life of 10 years.
- Salaries and wages of the boiler house:
 - 15 skilled workers @ ₹ 1,300 per month each
 - 30 semi-skilled workers @ ₹ 1,100 per month each
 - 50 unskilled workers @ ₹ 900 per month each
- Recovery on account of sale of ashes: 150 tonnes @ ₹ 60 per tonne.
- Salary and wages of generating station:
 - 60 skilled workers @ ₹ 1,200 per month each
 - 20 semi-skilled workers @ ₹ 1,000 per month each
 - 30 unskilled workers @ ₹ 850 per month each
- Repairs and maintenance of generating equipment: ₹ 30,000
- Depreciation of generating equipment: ₹ 75,000
- Insurance premium on generating plant: ₹ 15,000
- Fuel handling and storage expenses: ₹ 8,000
- Rent of generating station premises: ₹ 12,000
- Share of administration charges: ₹ 65,000
- Total number of units generated (before losses): 2,00,000 units
- Normal losses in the process: 3,000 units generated.

- Electricity purchased from grid to meet emergency requirements: 5,000 units @ ₹ 6 per unit.
- 10% of the effective units generated are used internally by the generator department itself

Additional Information:

- The purchased electricity units are not subject to any loss adjustment.
- Emergency purchase units should be added to the effective units available for use.

You are required to CALCULATE the total cost of electricity generated and the cost of generating electricity per unit.

Marginal Costing

16. N.P. Ltd. produces two products P and Q. The draft budget for the next month is as under:

	P	Q
Budgeted Production and Sales (units)	40,000	80,000
Selling Price ₹ / unit	25	50
Total Costs ₹ / unit	20	40
Machine Hours / unit	2	1
Maximum Sales Potential (units)	60,000	1,00,000

The fixed expenses are estimated at ₹ 9,60,000 per month. The company absorbs fixed overheads on the basis of machine hours which are fully utilised by the budgeted production and cannot be further increased.

When the budget was discussed, the Managing Director stated that the product mix should be altered to yield optimum profit.

The Marketing Director suggested that he would introduce a new Product-C, each unit of which will take 1.5 machine hours. However, a new machine involving a capital outlay of ₹ 2,00,000 is to be installed for processing Product-C. The additional fixed overheads relating to the

new machine was estimated at ₹ 60,000 per month. The variable cost of Product- C was estimated at ₹ 21 per Unit.

Required:

- (i) Calculate the profit as per draft budget for the next month.
- (ii) Revise the product mix based on data given for P and Q to yield optimum profit.
- (iii) The company decides to discontinue either Product- P or Q whichever is giving lower profit and proposes to substitute Product- C instead. Fix the selling price of product- C in such a way as to yield 15% return on additional capital employed besides maintaining the same overall profit as envisaged in (ii) above.

Budgets and budgetary control

17. X Ltd. has carefully analyzed market trends and consumer behaviour to project its sales figures for the upcoming year as follows:

Quarter	Sales (Units)
I	45,000
II	56,250
III	61,875
IV	67,500

Information regarding production and finished goods stock is as follows:	
Opening Stock of Finished Goods	11,250 Units
Expected Closing Stock of Finished Goods at the end of the year	18,375 Units
Production Pattern	75% of Current Quarter Sales
	Plus 25% of Next Quarter Sales
Closing Stock of current quarter to be maintained	25% of Next Quarter Sales

Further information regarding stock of raw materials is as follows:

Opening Stock of Raw Materials	15,000 Units
Required Closing Stock of Raw Materials at the end of the year	7,500 Units

Each unit of finished output requires 4 kg. of raw materials.

The company plans to acquire its total annual requirement of raw materials during the first three quarters, in the specified proportions and at the prices detailed below:

Quarter	Purchase of raw materials % to total annual requirement in quantity	Price per kg. (₹)
I	25%	12
II	50%	13
III	25%	14

You are required to PREPARE the following for the next year, quarter wise:

- (i) Production budget (in units).
- (ii) Raw material consumption budget (in quantity).
- (iii) Raw material purchase budget (in quantity).
- (iv) Raw material purchase budget (in value).

Miscellaneous

18. (a) Though Cost Accounting and Management Accounting are used synonymously, but there are a few differences. ENUMERATE any five differences.
- (b) A manufacturing company based in India is planning to transition from its traditional costing system to Activity-Based Costing (ABC) in order to improve cost accuracy and gain better insights into product profitability. Given the company's diverse product portfolio and the complexity of its production processes, the

implementation of ABC is expected to provide a more precise allocation of overheads.

You are required to SUGGEST the key practical stages involved in successfully implementing an Activity-Based Costing system.

- (c) Fixed Budget does not change with actual volume of activity achieved, however, Flexible Budget can be re-casted on the basis of activity level to be achieved. DISCUSS other differences between these two type of budget classified based on their capacity.
- (d) DESCRIBE cost unit and give EXAMPLE(S) of cost unit for the following Industry or Product:

Industry or Product
Construction
Transport
Technology
Oil
Gas
Electricity